Human-Centred AI Masters
December 2026
THES H6011 Research Thesis, 30 Credits

# Optimising Commercial Building Energy Consumption Using LSTM

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A thesis submitted in the partial fulfilment for the
**Masters in Science in Human-Centred Artificial Intelligence**

in the
Faculty of Computing, Digital and Data
School of Enterprise Computing and Digital Transformation

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# Declaration

I hereby declare that this dissertation is entirely my own work and has not been submitted, in whole or in part, for any other degree or professional qualification at this or any other institution. Where the work of others has been used, it has been appropriately cited and acknowledged within the text.

I acknowledge the use of Generative Artificial Intelligence (GenAI) tools in supporting the development of this report. These tools were used to assist with coding tasks, numerical calculations, data processing, and the refinement of written content. They also supported improvements in the clarity, structure, and professional presentation of the report.

All AI-generated outputs were critically reviewed, validated, and edited by the author. Full responsibility for the accuracy, interpretation, analysis, and final content of this dissertation remains solely with the author.

This declaration is provided in the interest of transparency regarding the use of AI-assisted tools during the preparation of this report.



Dikshitha Kalva Vinod
September 2, 2026

# Abstract

Commercial buildings generate large volumes of operational data through Internet of Things (IoT) sensors that continuously monitor electrical systems and energy consumption. Although high-frequency sensor data provides valuable opportunities for energy forecasting and management, the black-box nature of deep learning models can limit transparency and user understanding in real-world applications. This research investigates Long Short-Term Memory (LSTM) networks for short-term commercial building energy consumption forecasting using a real-world multivariate time-series IoT dataset provided by Electric Blue Technology. The dataset was collected from a commercial frozen food warehouse and contained 12,587 sensor observations recorded at approximately 15-minute intervals. Following preprocessing and data cleaning, 8,571 observations were retained for model development and evaluation. Six deep learning architectures designed for time-series forecasting, including Basic LSTM, Regularised LSTM, Stacked LSTM, Deep Stacked LSTM, Bidirectional LSTM (BiLSTM), and Gated Recurrent Unit (GRU) models, were developed and compared. The Stacked LSTM demonstrated strong forecasting performance and was further optimised through hyperparameter tuning. An ablation study was conducted to examine the effects of network architecture and regularisation. To improve transparency and interpretability, Explainable Artificial Intelligence (XAI) techniques, including SHAP, LIME, and Permutation Feature Importance, were integrated into the forecasting framework. These methods provided insights into model behaviour and identified influential predictors, supporting Human-Centred AI principles and trustworthy AI considerations reflected in ALTAI [1]. Model performance was evaluated using MAE, RMSE, MSE, sMAPE, and $R^2$. The results showed that the optimised Stacked LSTM achieved reliable forecasting performance, while XAI analyses consistently identified electrical current measurements as the most influential predictors of energy consumption. Overall, the proposed framework demonstrates how accurate deep learning forecasting can be combined with explainability to support transparent and informed energy-management decisions in commercial buildings.

### Keywords

LSTM, Energy Consumption Forecasting, Internet of Things (IoT), Explainable Artificial Intelligence (XAI), Human-Centred Artificial Intelligence (HCAI), Commercial Building Energy Management.

# Acknowledgement

I would like to express my sincere gratitude to my supervisor, Dr Na Li, and my co-supervisor, Dr Rajesh Jaiswal, Lecturers in the School of Enterprise Computing and Digital Transformation at TU Dublin, for their invaluable guidance, continuous support, and constructive feedback throughout this research. Their expertise, encouragement, and insightful recommendations were instrumental in shaping the direction, methodology, and successful completion of this dissertation.

I would also like to extend my sincere thanks to Paul Donohoe, Managing Director of Electric Blue Technology, for his industry guidance and for providing access to the real-world dataset used in this research. His support and practical insights contributed significantly to the relevance and applicability of this study.

I am grateful to Technological University Dublin and the School of Enterprise Computing and Digital Transformation for providing an excellent academic environment, resources, and opportunities that enabled the successful completion of this Master's programme. The knowledge and skills gained throughout the programme have been invaluable in undertaking this research.

Finally, I would like to acknowledge all the lecturers, staff, and members of TU Dublin who contributed to my learning experience and academic development during my studies. Their dedication and support have played an important role in my educational journey and in the completion of this dissertation.

# Contents

**List of Abbreviations** ii

**List of Figures** iii

**List of Tables** iv

**1 Introduction** 1
- 1.1 Background and Motivation . . . 1
- 1.2 Research Problem and HCAI Relevance . . . 1
- 1.3 Research Questions and Objectives . . . 1

**2 Literature Review** 3
- 2.1 Energy Datasets for Building Energy Analytics . . . 3
- 2.2 Models for Energy Consumption Forecasting . . . 4
- 2.3 Anomaly Detection in Building Energy Systems . . . 4
- 2.4 Explainable and Human-Centred Artificial Intelligence . . . 4
- 2.5 Research Gaps and Study Positioning . . . 5

**3 Data Collection and Preparation** 6
- 3.1 Data Collection . . . 6
  - 3.1.1 Data Description . . . 6
- 3.2 Data Preprocessing . . . 6
  - 3.2.1 Dataset Loading, JSON Expansion and Timestamp Conversion . . . 7
  - 3.2.2 Missing Value Analysis and Treatment . . . 7
- 3.3 Feature Engineering . . . 7
  - 3.3.1 Feature Analysis . . . 8
  - 3.3.2 Feature Selection Decisions . . . 8

**4 AI Modelling** 10
- 4.1 Model Comparison - Selection of Forecasting Models . . . 10
- 4.2 Hyperparameter Tuning . . . 12
- 4.3 Evaluation of the Optimised Stacked LSTM Model . . . 13
- 4.4 Ablation Study . . . 13
- 4.5 Explainable Artificial Intelligence . . . 13

**5 Results and Discussion** 14
- 5.1 Results . . . 14
  - 5.1.1 Best Performing Model . . . 14
  - 5.1.2 Forecasting Accuracy Results . . . 14
  - 5.1.3 SHAP Analysis . . . 14
  - 5.1.4 LIME Analysis . . . 15
  - 5.1.5 Permutation Feature Importance . . . 15

5.1.6 Comparison of Explainability Techniques . . . . . 16
5.2 Discussion . . . . . 16
5.2.1 Addressing Research Questions . . . . . 16
5.2.2 Human-Centred AI Perspective . . . . . 17
5.2.3 Practical Applications . . . . . 18
5.3 Self-Appraisal . . . . . 18

**6 Conclusion and Future Work** **19**
6.1 Conclusion . . . . . 19
6.2 Future Work . . . . . 20

**Bibliography** **22**

# List of Abbreviations

AI - Artificial Intelligence
ALTAI - Assessment List for Trustworthy Artificial Intelligence
ARIMA - AutoRegressive Integrated Moving Average
BiLSTM - Bidirectional Long Short-Term Memory
CNN - Convolutional Neural Network
CSV - Comma-Separated Values
DL - Deep Learning
GDPR - General Data Protection Regulation
GRU - Gated Recurrent Unit
HCAI - Human-Centred Artificial Intelligence
IoT - Internet of Things
LIME - Local Interpretable Model-Agnostic Explanations
LSTM - Long Short-Term Memory
MAE - Mean Absolute Error
MSE - Mean Squared Error
RMSE - Root Mean Squared Error
RNN - Recurrent Neural Network
SDG - Sustainable Development Goal
SHAP - SHapley Additive exPlanations
sMAPE - Symmetric Mean Absolute Percentage Error
SVR - Support Vector Regression
XAI - Explainable Artificial Intelligence

# List of Figures

1.1 Methodology . . . . . 2

3.1 Distribution of warehouse power consumption by hour of day . . . . . 8

4.1 Basic LSTM Loss and MAE Curves . . . . . 10
4.2 Regularised LSTM Loss and MAE Curves . . . . . 11
4.3 Stacked LSTM Loss and MAE Curves . . . . . 11

5.1 SHAP Feature Importance . . . . . 15
5.2 LIME Feature Importance . . . . . 15
5.3 Permutation Feature Importance . . . . . 16

# List of Tables


2.1 Comparison of datasets used in previous studies... . . . . 3
2.2 Comparison of forecasting models used for energy prediction . . . . . 5

3.1 Dataset Reduction During Preprocessing . . . . . 7
3.2 Selected Feature Groups Used for Model Development . . . . . 9

4.1 Model Architecture and Hyperparameter Configurations . . . . . 10
4.2 Performance Comparison of LSTM and GRU Models . . . . . 11
4.3 LSTM Hyperparameter Configurations . . . . . 12
4.4 Evaluation Metrics for Different LSTM Configurations . . . . . 12
4.5 Ablation Study Results . . . . . 13

5.1 Feature importance and ranking obtained from SHAP, LIME, and Permutation Importance. Values are reported as Importance (%). Rank. . 16

# 1 Introduction

## 1.1 Background and Motivation

Commercial buildings generate large volumes of operational data through Internet of Things (IoT) sensors that continuously monitor electrical systems and equipment. This data creates opportunities to apply Artificial Intelligence (AI) techniques to improve energy management and support operational decision-making [2, 3]. One important application is short-term energy forecasting, which allows organisations to predict future energy demand and identify consumption patterns. As smart building technologies become more widespread, developing accurate forecasting models for high-frequency sensor data has become an important research area [4, 5].

Despite the availability of large amounts of operational data, forecasting energy consumption accurately remains challenging because energy usage contains complex temporal and non-linear patterns that traditional forecasting methods often fail to capture. Deep Learning (DL) models, particularly Long Short-Term Memory (LSTM) networks, have shown strong performance in time-series forecasting by learning patterns and dependencies from historical data [6, 7, 8]. However, LSTM models often operate as black-box systems, making it difficult for users to understand how predictions are generated. This lack of interpretability can reduce trust in AI-driven decisions and limit the adoption of forecasting systems in real-world commercial building operations [9].

## 1.2 Research Problem and HCAI Relevance

Although DL models can produce accurate forecasts, their decision-making processes are often difficult to interpret, particularly when large operational datasets are involved. Explainable Artificial Intelligence (XAI) techniques address this limitation by revealing how input features influence model predictions, thereby enhancing transparency, supporting informed decision-making, and increasing confidence in AI-assisted forecasting systems [9, 10].

This research is grounded in the Human-Centred AI principles of transparency and interpretability. Transparency makes model reasoning visible, while interpretability helps users understand how input variables influence forecasting outputs. These principles are addressed through Explainable Artificial Intelligence (XAI) techniques, including SHAP, LIME, and Permutation Feature Importance [2, 3].

## 1.3 Research Questions and Objectives

The research questions are:

- **RQ1:** To what extent can an AI model predict short-term commercial building energy consumption using high-resolution IoT sensor data?

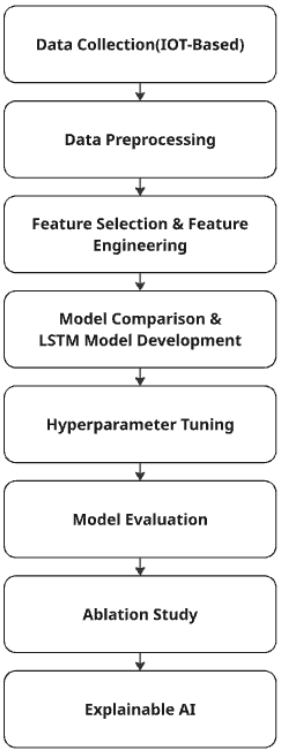



Figure 1.1: Methodology

- **RQ2:** How can XAI techniques, such as SHAP and LIME, improve the transparency of LSTM-based forecasting models and identify the key features contributing to energy-consumption predictions?

This research aims to develop and evaluate an LSTM-based framework for predicting short-term energy consumption in commercial buildings using high-resolution IoT sensor data, and to investigate how XAI techniques can improve model transparency and identify the features contributing to energy-consumption predictions. IoT sensor data collected from a commercial warehouse facility are analysed, and an LSTM forecasting model is developed and evaluated using standard performance metrics. SHAP, LIME, and permutation feature importance are applied to explain model predictions and identify the most influential input variables.

# 2 Literature Review

## 2.1 Energy Datasets for Building Energy Analytics

The growth of Internet of Things (IoT) technologies has enabled continuous collection of high-resolution energy data from commercial buildings, including current, voltage, and real power measurements. IoT-enabled platforms support energy monitoring, forecasting, and operational decision-making. For example, Energy Elephant[1] provides energy monitoring and sustainability reporting, Wattics[2] offers energy analytics and forecasting, while GridBeyond[3] provides intelligent energy management solutions. These applications demonstrate the value of sensor-based data for energy optimisation.

A comparison of relevant datasets is presented in Table 2.1. Existing datasets combine electrical, environmental, and operational variables. The Electric Blue dataset used in this research provides detailed electrical measurements from a real-world commercial facility, including current, voltage, active power, reactive power, power factor, and apparent power. This makes it suitable for energy forecasting, anomaly detection, and Explainable Artificial Intelligence (XAI). Unlike datasets focused mainly on environmental or occupancy variables, it provides detailed electrical information from an operational frozen food warehouse.

| **Dataset** | **Building Type** | **Key Features** | **Target Variable** |
|---|---|---|---|
| Tetouan City Dataset [6] | Residential Building | Temperature, Humidity, Zone Power Consumption | Energy Consumption |
| DECODE Dataset [8] | Commercial Building | Weather Data, Occupancy, Historical Load | Energy Demand |
| UDLA Ecuador Dataset [10] | University Building | Active Power, Reactive Power, Power Factor | Power Consumption |
| OpenEI Hospital Dataset [11] | Hospital Building | Load Profiles, Time Variables, Historical Usage | Electricity Demand |
| Energy Efficiency Dataset [12] | Commercial Building | Building Characteristics, HVAC Load, Heating Load | Energy Consumption |
| Hospitality Building Dataset [13] | Hotel Building | Energy Usage, Operational Variables | Electricity Demand |
| IoT Smart Building Dataset [14] | Smart Commercial Building | IoT Sensors, HVAC Data, Temperature, Energy Usage | Building Energy Consumption |
| Cooling Load Dataset [15] | Commercial Building | Indoor Conditions, Outdoor Conditions, Cooling Load | Cooling Energy Demand |
| **Electric Blue Dataset (Present Study)** | **Frozen Food Warehouse** | **Current, Voltage, Active Power, Reactive Power, Power Factor, Apparent Power** | **Energy Consumption** |

Table 2.1: Comparison of datasets used in previous studies...

[1] https://www.energyelephant.com/
[2] https://wattics.com/
[3] https://gridbeyond.com/

## 2.2 Models for Energy Consumption Forecasting

Long Short-Term Memory (LSTM) networks are widely used for energy forecasting because they can capture short- and long-term temporal dependencies and non-linear relationships in sequential data. Studies have shown that LSTM can outperform traditional approaches such as Linear Regression, Random Forest, and Support Vector Regression (SVR) in energy prediction tasks [8, 12, 7]. This makes LSTM suitable for building energy consumption, where demand is influenced by operational schedules, equipment behaviour, and historical patterns.

Alternative approaches include Gated Recurrent Units (GRU), Recurrent Neural Networks (RNN), and Autoregressive Integrated Moving Average (ARIMA) [10, 16, 13]. GRU generally requires fewer computational resources than LSTM, while RNN can model sequential relationships but may suffer from vanishing gradients. ARIMA is effective for trends and seasonal patterns but is less suitable for highly dynamic and non-linear energy data (Table 2.2).

More advanced architectures, including Bidirectional LSTM (BiLSTM), CNN-LSTM, and Attention-LSTM, can capture richer temporal patterns [6, 14, 17, 16, 18]. However, these approaches increase model complexity and computational requirements. Standard LSTM was therefore selected for this research because it provides a suitable balance between forecasting performance, computational efficiency, and implementation simplicity [19, 13]. Although the dataset lacks environmental variables such as temperature and humidity, its high-resolution electrical measurements provide sufficient temporal information for sequence-based forecasting.

## 2.3 Anomaly Detection in Building Energy Systems

Anomaly detection is important for identifying inefficiencies and abnormal behaviour in building energy systems. A common prediction-based approach compares predicted and actual energy consumption and identifies significant deviations as potential anomalies [11]. However, deviations may also result from seasonal effects, occupancy changes, or operational variations, which can lead to false-positive detections [20, 21]. Therefore, prediction-based anomaly detection should be interpreted alongside the operational context of the building. The strategy adopted in this research follows this approach while recognising these limitations.

## 2.4 Explainable and Human-Centred Artificial Intelligence

Human-Centred Artificial Intelligence emphasises transparency, explainability, and interpretability in AI-assisted decision-making. This is particularly relevant to building energy management, where facility managers may need to understand the factors influencing forecasts before using them for operational decisions. Explainable Artificial Intelligence (XAI) addresses this challenge by providing insights into model behaviour and feature contributions [9, 10].

Techniques such as SHAP, LIME, and permutation feature importance can identify variables that influence model predictions, improving transparency. Ethical frameworks, including the Assessment List for Trustworthy Artificial Intelligence (ALTAI) [1], also emphasise transparency and responsible AI deployment [3, 4, 5, 2, 15]. Integrating XAI with forecasting can therefore help bridge the gap between predictive performance and practical human-centred use.

Within ALTAI [1], transparency is particularly relevant to forecasting systems. Transparency requires that the basis of a model's outputs be understandable and

communicable to users. Explainability techniques such as SHAP, LIME, and Permutation Feature Importance support transparency by revealing how input variables influence model predictions. For this reason, transparency forms the primary ALTAI focus of this research.[3, 4].

## 2.5 Research Gaps and Study Positioning

Despite advances in building energy forecasting, several research gaps remain. Many studies prioritise predictive accuracy while providing limited interpretability and decision support for end users [9, 10]. This lack of transparency may reduce stakeholder trust and hinder the adoption of AI-based energy management systems.

In addition, many forecasting approaches are evaluated using benchmark or simulated datasets rather than operational data collected from real-world commercial environments [4, 5]. Existing research also frequently focuses on environmental, occupancy, or limited electrical variables, with comparatively less attention given to detailed measurements such as current, voltage, reactive power, apparent power, and power factor [10, 12].

To address these gaps, this research develops an LSTM-based forecasting framework using real-world IoT data collected from a commercial frozen food warehouse. The study combines detailed electrical measurements with XAI techniques to investigate the variables influencing energy consumption forecasts. By integrating forecasting with explainability, the research aims to contribute towards more transparent, interpretable, and practically deployable AI solutions for commercial building energy management.

| **Model** | **Description** | **Strengths** | **Limitations** |
|---|---|---|---|
| Linear Regression | Linear relationship between inputs and target. | Simple and interpretable. | Limited for non-linear patterns. |
| ARIMA | Statistical time-series forecasting model. | Effective for trends and seasonality. | Less effective for complex non-linear data. |
| SVR | Machine learning regression using support vectors. | Good for small, non-linear datasets. | Limited temporal modelling. |
| Random Forest | Ensemble of decision trees. | Handles non-linear relationships. | Does not directly model temporal dependencies. |
| RNN | Recurrent model for sequential data. | Learns temporal relationships. | Prone to vanishing gradients. |
| GRU | Simplified gated recurrent network. | Faster and less computationally expensive than LSTM. | Less effective for very long dependencies. |
| LSTM | Gated recurrent network for sequential data. | Captures short- and long-term dependencies. | Higher computational requirements. |
| BiLSTM | LSTM processing sequences in both directions. | Captures richer temporal context. | More complex and computationally expensive. |
| CNN-LSTM | Combines CNN feature extraction with LSTM. | Captures features and temporal patterns. | Higher complexity and computational cost. |
| Attention-LSTM | LSTM with an attention mechanism. | Focuses on important inputs and time steps. | More complex and less interpretable. |

Table 2.2: Comparison of forecasting models used for energy prediction

# 3 Data Collection and Preparation

## 3.1 Data Collection

This study utilised an industrial Internet of Things (IoT) dataset provided by Electric Blue Technology. The dataset was collected from a commercial frozen food warehouse using IoT sensors to monitor electrical energy consumption, enabling the analysis of real-world warehouse energy usage patterns [2, 3].

The dataset was provided directly by the industry partner, Electric Blue Technology, under an agreement permitting its use for academic research purposes. The dataset consisted solely of electrical and device-monitoring measurements collected from warehouse equipment and did not contain any personal or personally identifiable information relating to individuals. As a result, no data belonging to, or capable of identifying, any individual was collected, processed, or stored at any stage of this research, and the dataset contained only electrical and device-monitoring measurements and did not include personal or personally identifiable information. Consequently, the research did not involve the processing of personal data as defined within the scope of the General Data Protection Regulation (GDPR) [22].

### 3.1.1 Data Description

The dataset consisted of 12,587 observations recorded at approximately 15-minute intervals and was structured as a multivariate time-series dataset suitable for energy forecasting. Initially, the dataset contained seven fields: `tsmeasuredat`, `tenant`, `deviceid`, `location`, `payload`, `meta`, and `ts_ingested_db`. The `payload` field contained sensor measurements stored in JSON format, while the `meta` field contained device information [4, 5].

To support analysis, the JSON payload was expanded into individual variables, increasing the dataset from seven to twenty columns. The resulting variables included electrical measurements such as `current_chan1`, `current_chan2`, `current_chan3`, `current_chan4`, `assumed_voltage_v`, `real_power_total_kW`, and `assumed_power_factor`, together with device-monitoring variables including battery level and signal strength.

The target variable for this study was `real_power_total_kW`, representing total warehouse electrical power consumption in kilowatts (kW). The remaining variables were used as predictors during feature analysis and model development. The high-frequency IoT measurements provided a suitable foundation for short-term forecasting and explainability analysis.

## 3.2 Data Preprocessing

Data preprocessing converted the raw IoT data into a format suitable for forecasting analysis. The process included dataset inspection, JSON expansion, timestamp conversion, duplicate checking, missing-value treatment, and creation of a cleaned dataset. No duplicate records or timestamps were identified [2, 4].

The preprocessing stage was essential because deep learning models are highly sensitive to data quality issues. Raw IoT sensor data often contain missing observations, inconsistencies, and measurement artefacts resulting from communication interruptions, sensor maintenance, or temporary device failures[2, 4]. If left untreated, these issues can negatively affect model training and reduce forecasting accuracy. Consequently, a systematic preprocessing approach was adopted to ensure that the resulting dataset accurately reflected actual warehouse energy consumption behaviour. The preprocessing decisions were designed to preserve temporal continuity while minimising the loss of useful information required for forecasting and explainability analysis.

Because the dataset represents chronological time-series data, the training, validation, and test partitions were created while preserving temporal order. This approach reduces the risk of data leakage and ensures that the forecasting models are evaluated on future observations that were not available during training.

### 3.2.1 Dataset Loading, JSON Expansion and Timestamp Conversion

The raw CSV file contained 12,587 records across seven columns. Following extraction of the JSON-formatted payload data, the dataset expanded to twenty columns and the original payload and metadata fields were removed. All timestamps were converted to a standard datetime format and sorted chronologically. The original dataset covered the period from 30 January 2026 to 21 July 2026.

### 3.2.2 Missing Value Analysis and Treatment

Missing-value analysis identified 4,016 missing values in `assumed_voltage_v`, `real_power_total_kW`, and `assumed_power_factor`, while `assumed_noise_floor_a` contained 11,835

missing values. Records containing missing target values were removed, resulting in the exclusion of 4,016 observations.

Missing values in the remaining numerical variables were handled using interpolation followed by forward- and backward-fill techniques. After preprocessing, no missing values remained within the dataset [6, 8]. The final cleaned dataset contained 8,571 observations and covered the period from 30 January 2026 to 9 June 2026.

| **Preprocessing Stage** | **Observations** |
|---|---|
| Raw Dataset | 12,587 |
| Removed Missing Target Values | 4,016 |
| Final Clean Dataset | 8,571 |

Table 3.1: Dataset Reduction During Preprocessing

## 3.3 Feature Engineering

Feature understanding was conducted to identify the variables most relevant to predicting energy consumption. Statistical analysis, feature-importance methods, and temporal analysis were used to support feature selection [10, 7, 12].

Following preprocessing, the dataset contained electrical measurements, device-monitoring variables, and temporal features. The principal electrical variables included `current_chan1`, `current_chan2`, `current_chan3`, `current_chan4`, `assumed_voltage_v`, and `assumed_power_factor`

, while battery level and signal strength provided device-monitoring information. Temporal features derived from timestamps included hour, day of week, and cyclical representations of time.

### 3.3.1 Feature Analysis

Multiple feature-selection approaches were applied to identify the most influential predictors of energy consumption. Correlation analysis showed very strong positive relationships between the target variable and `current_chan1` (0.9968), `current_chan2` (0.9963), and `current_chan3` (0.9963). Other variables displayed substantially weaker relationships [9, 10].

Feature-importance analysis produced consistent findings. Random Forest importance identified `current_chan3`, `current_chan1`, and `current_chan2` as the dominant predictors, while permutation importance showed that these three variables accounted for more than 90% of overall predictive importance. Temporal variables, particularly `hour_sin` and `hour_cos`, also contributed useful information by capturing daily consumption cycles.

Standard deviation analysis further confirmed the importance of the current-channel variables, which showed the highest variation in the dataset. In contrast, battery level, power factor, and several device-related variables displayed relatively limited variability and predictive value.

Temporal analysis identified recurring consumption patterns throughout the day, supporting the inclusion of time-based features. Figure 3.1 presents the distribution of warehouse power consumption across the 24-hour cycle, illustrating a clear operational pattern of elevated and more variable load during daytime working hours and reduced, more stable load overnight. This pattern visually supports the inclusion of the `hour`, `hour_sin`, and `hour_cos` features described above and reinforces the correlation-based evidence for their predictive relevance. Weekly analysis showed only minor differences between weekday and weekend energy consumption, but sufficient variation existed to justify retaining day-of-week and weekend indicators.

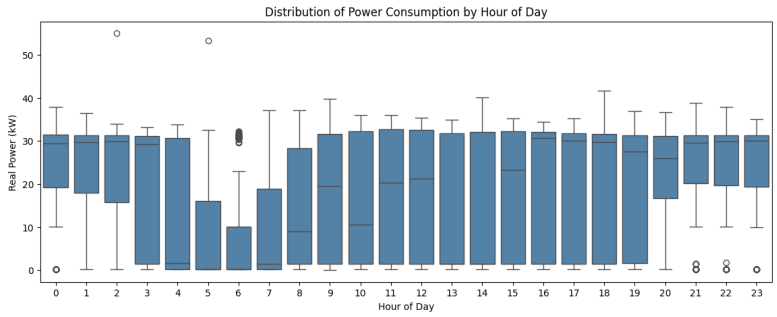



Figure 3.1: Distribution of warehouse power consumption by hour of day

### 3.3.2 Feature Selection Decisions

Feature selection was based on the combined results of correlation analysis, feature importance methods, temporal analysis, and domain knowledge. Domain expertise provided by the industry partner informed the interpretation of sensor variables and their expected relationships with energy consumption, ensuring that feature-selection decisions were grounded in both statistical evidence and operational understanding. This ensured that feature selection was supported by both data-driven evidence and real-world operational understanding. Table 3.2 summarises the final features retained for model development [8, 13].

The variables `current_chan1`, `current_chan2`, and `current_chan3` were consistently identified as the most influential predictors of energy consumption. Temporal features were retained to capture recurring consumption patterns, while `assumed_noise_floor_a` was removed because of extensive missing values and limited analytical value (Table 3.2). Identifier variables such as IMEI, IMSI, tenant, device identifier, location, and model information were excluded because they remained constant throughout the dataset [7, 14].

In addition to predictor importance, feature analysis was used to identify potential redundancy between variables. Several electrical measurements provided related information regarding overall power consumption because they were derived from the same physical electrical system. However, variables were not excluded solely based on correlation, as some features may contribute complementary information when processed by deep learning models. Retaining a combination of electrical and temporal variables ensured that the forecasting model could capture both instantaneous electrical demand and recurring behavioural patterns over time. This approach supported the development of a balanced feature set while avoiding unnecessary complexity. Furthermore, insights provided by the industry expert who supplied the dataset were used to validate the practical relevance of the selected features and their relationships with energy consumption. This ensured that feature selection was informed not only by statistical evidence but also by real-world operational understanding of the commercial building environment.

| **Feature Group** | **Features** |
|---|---|
| Current | `current_chan1-4` |
| Electrical | `assumed_voltage_v`, `assumed_power_factor` |
| Device status | `battery`, `signal` |
| Time of day | `hour`, `hour_sin`, `hour_cos` |
| Calendar | `dayofweek`, `weekend` |

Table 3.2: Selected Feature Groups Used for Model Development

# 4 AI Modelling

## 4.1 Model Comparison - Selection of Forecasting Models

This study evaluated six recurrent deep learning architectures for short-term commercial building energy forecasting: Basic LSTM, Regularised LSTM, Stacked LSTM, Deep Stacked LSTM, Bidirectional LSTM (BiLSTM), and Gated Recurrent Unit (GRU) (Table 4.1), enabling the impact of model depth, regularisation, and recurrent structure to be assessed [6, 7, 8]. All models were trained on the same dataset and evaluated using MAE, RMSE, MSE, sMAPE, and $R^2$ [10, 13, 14].

| Model | LSTM Layers | Units | Dropout |
|---|---|---|---|
| Basic LSTM | 1 | 32 | None |
| Regularised LSTM | 1 | 64 | 0.20 |
| Stacked LSTM | 2 | 64 → 32 | 0.20 |
| Deep Stacked LSTM | 3 | 128 → 64 → 32 | 0.20 |
| BiLSTM | 1 | 64 | 0.20 |
| GRU | 1 | 64 | 0.20 |

Table 4.1: Model Architecture and Hyperparameter Configurations

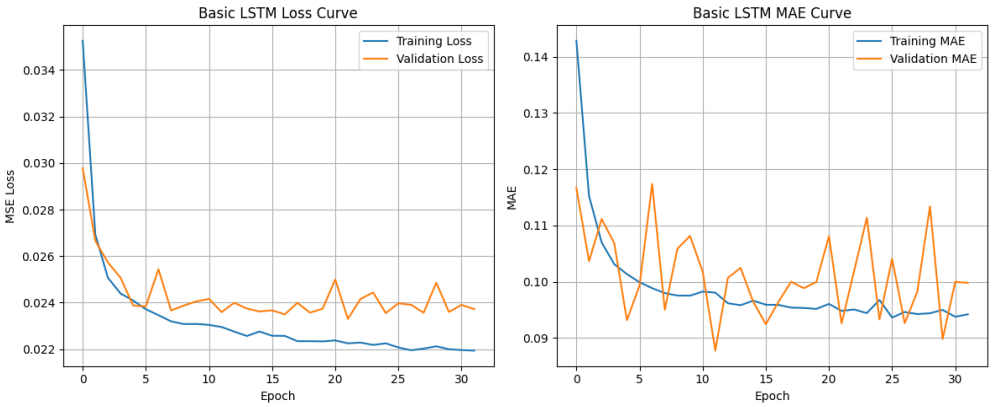



Figure 4.1: Basic LSTM Loss and MAE Curves

On test-set metrics alone, the Basic LSTM performed best (RMSE 7.6054, $R^2$ 0.6224), closely followed by BiLSTM (RMSE 7.6295, $R^2$ 0.6201); the Regularised LSTM and GRU were slightly weaker, and the Stacked and Deep Stacked LSTM recorded the highest RMSE values.

However, selection was not based on test-set accuracy alone. Training and validation curves (Figures 4.1-4.3) were examined for convergence and generalisation stability. The Basic LSTM's validation loss never stabilised, oscillating between 0.0233 and 0.0254 over its final 25 epochs against a training loss of 0.0219, with validation MAE fluctuating

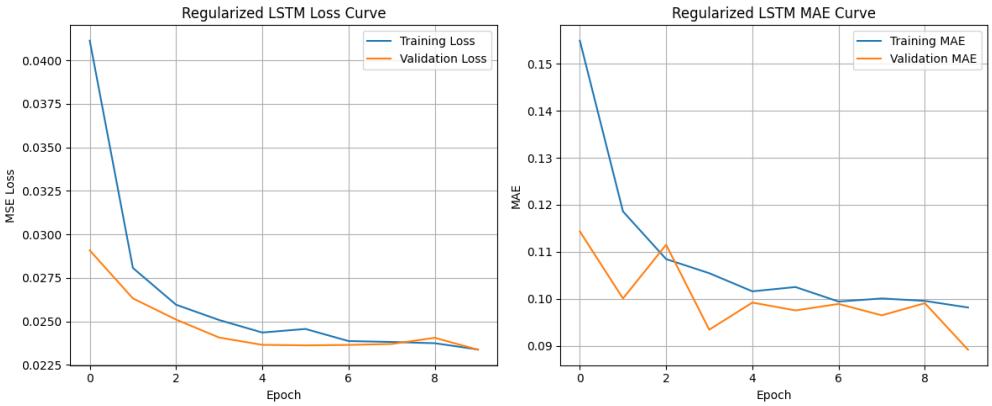



Figure 4.2: Regularised LSTM Loss and MAE Curves

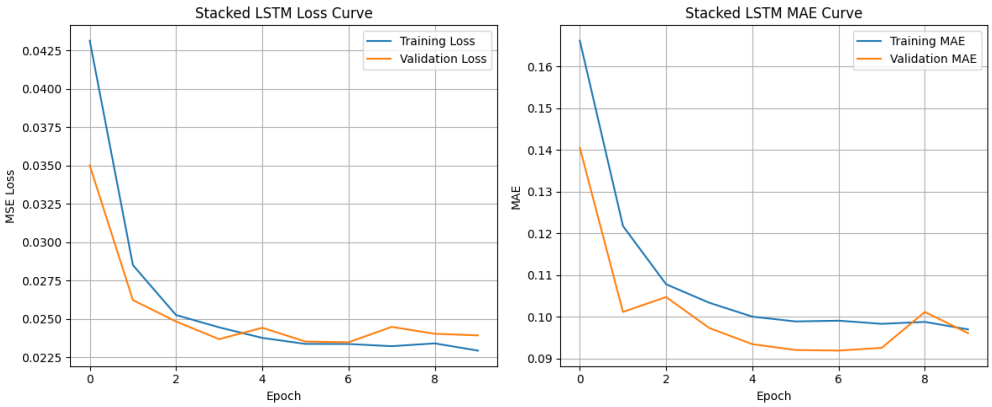



Figure 4.3: Stacked LSTM Loss and MAE Curves

| **Model** | **MSE** | **RMSE** | **MAE** | **$R^2$** | **sMAPE** |
|---|---|---|---|---|---|
| Basic LSTM | 57.8425 | 7.6054 | 5.3610 | 0.6224 | 49.2641 |
| Regularised LSTM | 67.1282 | 8.1932 | 5.6151 | 0.5618 | 51.0797 |
| Stacked LSTM | 92.9654 | 9.6419 | 7.4891 | 0.3932 | 57.4091 |
| Deep Stacked LSTM | 125.7974 | 11.2159 | 9.2699 | 0.1789 | 69.4701 |
| BiLSTM | 58.2086 | 7.6295 | 5.1198 | 0.6201 | 49.6965 |
| GRU | 68.8085 | 8.2951 | 6.0036 | 0.5509 | 52.2853 |

Table 4.2: Performance Comparison of LSTM and GRU Models

between 0.087 and 0.117. By contrast, the Stacked LSTM converged within 10 epochs, with training and validation losses tracking closely from epoch 4 and a final train-validation gap of just 0.0009 - roughly half the Basic LSTM's gap, reached in under a third of the epochs. The Deep Stacked LSTM performed considerably worse across all metrics (RMSE 11.2159, $R^2$ 0.1789), indicating that depth beyond two layers did not help.

These results from Table 4.2 show that test-set accuracy alone would have favoured the Basic LSTM, but this masked a less stable training process. The Stacked LSTM was therefore selected for further optimisation: comparable test-set performance, faster convergence, a smaller generalisation gap, and greater representational capacity than a single-layer model, a deliberate trade-off between raw accuracy and training reliability rather than a single-metric choice.

## 4.2 Hyperparameter Tuning

The Stacked LSTM was tuned across LSTM units, dropout rate, and batch size, using RMSE as the primary metric, supported by MAE, sMAPE, and $R^2$ [6, 17]. The baseline (64→32 units, dropout 0.2, batch size 32) was compared against seven alternative configurations (Table 4.3).

| Experiment ID | LSTM Layer 1 | LSTM Layer 2 | Dropout Rate | Batch Size |
|---|---|---|---|---|
| **Baseline Configuration** | **64** | **32** | **0.2** | **32** |
| H1 - Smaller Network | 32 | 16 | 0.2 | 32 |
| H2 - Larger Network | 128 | 64 | 0.2 | 32 |
| H3 - Equal Units | 64 | 64 | 0.2 | 32 |
| H4 - Low Dropout | 64 | 32 | 0.1 | 32 |
| H5 - High Dropout | 64 | 32 | 0.3 | 32 |
| H6 - Small Batch Size | 64 | 32 | 0.2 | 16 |
| H7 - Large Batch Size | 64 | 32 | 0.2 | 64 |

Table 4.3: LSTM Hyperparameter Configurations

| Experiment | MSE | RMSE | MAE | $R^2$ | sMAPE |
|---|---|---|---|---|---|
| Baseline Configuration | **57.9755** | **7.6142** | 5.0484 | **0.6216** | **48.7337** |
| H1 - Smaller Network | 58.9867 | 7.6803 | **4.9967** | 0.6150 | 49.1925 |
| H2 - Larger Network | 58.9744 | 7.6795 | 5.1403 | 0.6151 | 49.7551 |
| H3 - Equal Units | 62.6475 | 7.9150 | 5.6951 | 0.5911 | 51.5152 |
| H4 - Low Dropout | 61.7565 | 7.8585 | 5.1910 | 0.5969 | 49.3511 |
| H5 - High Dropout | 59.4471 | 7.7102 | 5.2504 | 0.6120 | 50.4055 |
| H6 - Small Batch Size | 62.0095 | 7.8746 | 5.6625 | 0.5952 | 51.3836 |
| H7 - Large Batch Size | 64.3191 | 8.0199 | 5.6180 | 0.5802 | 49.8845 |

Table 4.4: Evaluation Metrics for Different LSTM Configurations

The baseline achieved the lowest RMSE, lowest sMAPE, and highest $R^2$; only MAE was marginally lower for the smaller network. Larger networks and altered dropout or batch sizes all produced weaker results, indicating that additional capacity did not improve forecasting performance [8, 16]. Random seeds were fixed in NumPy, TensorFlow, and Python's `random` module to ensure reproducibility.

## 4.3 Evaluation of the Optimised Stacked LSTM Model

The final model, two LSTM layers (64, 32 units), dropout 0.2, batch size 32, learning rate 0.01 - achieved RMSE 7.6142 kW, MAE 5.0484 kW, sMAPE 48.73%, and $R^2$ 0.6216, explaining approximately 62% of the variation in warehouse energy consumption (Table 4.4) [7, 8, 10]. The unexplained variation may partly reflect the absence of environmental variables such as temperature and humidity, but the electrical and temporal features available provided a reliable foundation for the subsequent explainability analysis [6, 7, 13].

## 4.4 Ablation Study

An ablation study examined the effect of network complexity and dropout, comparing the Stacked LSTM against simplified and single-layer variants and models without dropout [19, 17], all trained on a normalised target (0-1 scale).

| Configuration | MSE | RMSE | MAE | $R^2$ | sMAPE |
|---|---|---|---|---|---|
| Small Model | 0.0191 | 0.1382 | 0.0928 | 0.6239 | 76.88 |
| Single LSTM 32 | 0.0192 | 0.1385 | 0.0886 | 0.6221 | 79.14 |
| Single LSTM 64 | 0.0192 | 0.1386 | 0.0943 | 0.6214 | 77.91 |
| No Dropout 1 | 0.0193 | 0.1391 | 0.0983 | 0.6189 | 76.15 |
| No Dropout 2 | 0.0200 | 0.1413 | 0.1043 | 0.6068 | 77.32 |
| No Dropout | 0.0200 | 0.1415 | 0.0959 | 0.6052 | 82.31 |
| Full Model | 0.0207 | 0.1438 | 0.1082 | 0.5926 | 78.27 |

Table 4.5: Ablation Study Results

Simpler architectures matched or slightly outperformed the full model: the Small Model achieved the lowest RMSE (0.1382) and highest $R^2$ (0.6239), and Single LSTM 32 the lowest MAE (0.0886). This does not contradict the earlier model selection the ablation used a normalised target, a different train-validation split, and fewer epochs, and the $R^2$ range across all eight configurations (0.593-0.624) was narrow (Table 4.5). The study's purpose was not to find a new optimal architecture but to test whether added depth or capacity provided real benefit once an appropriate architecture had been chosen [19]. Additional layers and units gave no clear improvement, while removing dropout consistently reduced performance, supporting dropout's regularising role [17].

The relatively high sMAPE values (49-83%) reflect its sensitivity when actual demand approaches zero, as during overnight hours, where small absolute errors produce large percentage errors [6, 16]. RMSE, MAE, and $R^2$ are less sensitive to these near-zero observations and were therefore treated as the primary indicators of forecasting quality, with sMAPE reported alongside rather than in isolation.

## 4.5 Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) was incorporated to address the black-box nature of deep learning models, since users need to understand the factors driving predictions before relying on them operationally. Three techniques were applied: SHAP, LIME, and Permutation Feature Importance, providing insight into the contribution of each input variable. Beyond improving transparency and user trust, XAI verified that the model's performance was driven by genuinely important features rather than spurious variables, supporting both interpretability and Human-Centred AI principles [9, 10].

# 5 Results and Discussion

## 5.1 Results

### 5.1.1 Best Performing Model

The optimised Stacked LSTM model was selected as the final forecasting framework following the model comparison and hyperparameter tuning experiments presented in Chapter 4. The final architecture consisted of two LSTM layers with 64 and 32 units, respectively, together with dropout regularisation and optimised training parameters. The model demonstrated stable learning behaviour throughout training and validation, indicating good generalisation to unseen data. These results suggest that the architecture was capable of capturing the temporal relationships present within the warehouse energy consumption dataset while maintaining a balance between forecasting accuracy and model robustness.

### 5.1.2 Forecasting Accuracy Results

The forecasting results indicate that the final Stacked LSTM model successfully captured meaningful temporal dependencies within the IoT sensor data and generated reliable short-term energy-consumption forecasts. As reported in Section 4.3, the model explained approximately 62% of the variability in warehouse energy consumption using the selected input features and a 24-hour historical observation window, demonstrating the effectiveness of sequence-based deep learning approaches for building energy forecasting [6, 7, 8, 13]. Overall, the results demonstrate the suitability of LSTM-based architectures for modelling commercial warehouse energy demand. The achieved forecasting performance suggests that high-frequency electrical measurements contain sufficient information to capture recurring consumption patterns and support short-term forecasting. Although prediction errors remained unavoidable due to the dynamic nature of energy consumption behaviour, the model consistently produced accurate forecasts and provided a reliable foundation for the explainability analyses discussed in the following sections.

### 5.1.3 SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to the Stacked LSTM to understand how individual features contributed to the forecasting model's predictions. The analysis revealed that the model relied primarily on the electrical current measurements (Figure 5.1), with `current_chan2`, `current_chan3`, and `current_chan1` identified as the most influential predictors. These variables represent the electrical load measured across the warehouse monitoring channels and therefore have a direct relationship with power consumption. Temporal features, including `hour_cos`, `hour_sin`, and `hour`, also contributed to model predictions by capturing daily operational cycles. The results suggest that the model learned meaningful relationships between electrical demand patterns and energy consumption, supporting both forecasting accuracy and model interpretability [9].

The dominance of current-channel measurements is physically meaningful because electrical current is directly associated with power consumption within electrical systems. This finding provides domain-level validation that the forecasting model is capturing physically meaningful relationships rather than relying on spurious correlations within the dataset.

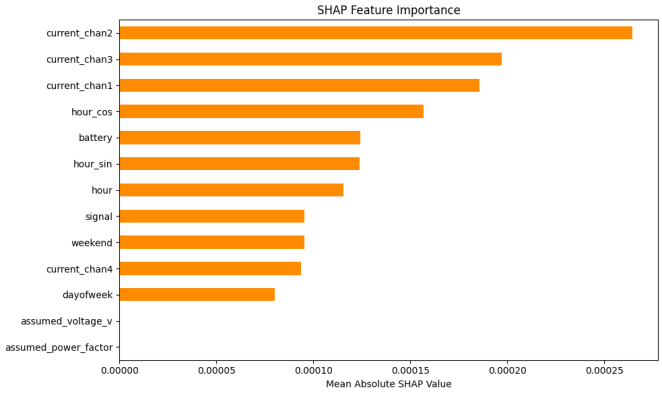



Figure 5.1: SHAP Feature Importance

### 5.1.4 LIME Analysis

LIME (Local Interpretable Model-Agnostic Explanations) was applied to explain individual forecasting predictions generated by the optimised Stacked LSTM model. The analysis identified `current_chan3`, `current_chan2`, and `current_chan1` as the most influential features affecting local prediction outcomes. In addition, temporal variables such as `hour_sin`, `hour`, `hour_cos`, `dayofweek`, and `weekend` contributed (Figure 5.2) to the model's decision-making process by capturing daily and weekly consumption patterns. The results indicate that the model combines electrical load measurements with temporal information to generate forecasts, demonstrating that both operational electricity demand and recurring usage patterns influence energy consumption predictions [17].

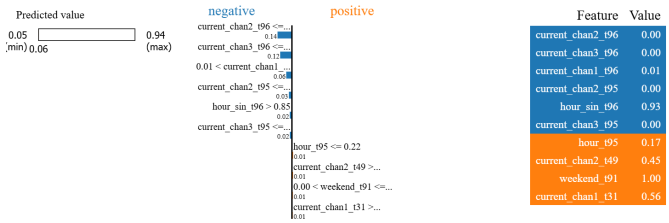



Figure 5.2: LIME Feature Importance

### 5.1.5 Permutation Feature Importance

Permutation Feature Importance was applied to evaluate the contribution of each feature by measuring the change in forecasting performance when individual variables were randomly shuffled. The analysis showed that `current_chan1`, `current_chan2`, and `current_chan3` were the most influential predictors, indicating that the forecasting model relied heavily on electrical current measurements. Temporal features, including `hour_sin`, `hour_cos`, and `weekend`, also contributed to prediction accuracy by capturing recurring energy consumption patterns. In contrast, variables such as `battery`, `assumed_voltage_v`, and `assumed_power_factor` had minimal influence (Figure 5.3). These findings confirm

that both electrical load measurements and temporal behaviour are important drivers of energy consumption forecasts [9, 10].

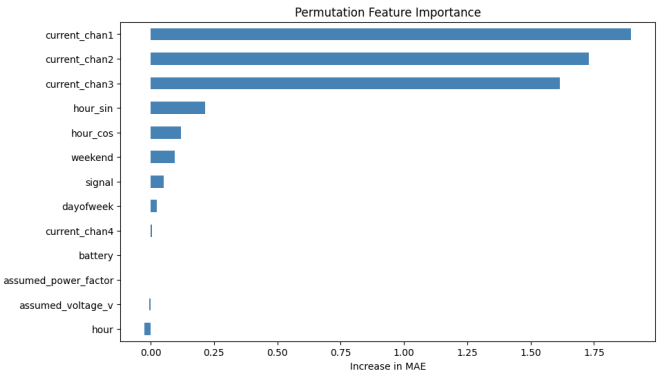



Figure 5.3: Permutation Feature Importance

### 5.1.6 Comparison of Explainability Techniques

A high level of consistency was observed across the SHAP, LIME, and Permutation Feature Importance analyses. As shown in Table 5.1, all three explainability techniques identified the current sensor measurements, namely `current_chan1`, `current_chan2`, and `current_chan3` as the most influential features driving energy consumption predictions. This agreement between pre-modelling feature analysis and post-modelling explainability results suggests that the LSTM model learned meaningful and physically relevant relationships within the dataset rather than relying on spurious correlations [9, 17]. Consequently, the consistency across multiple analytical methods increases confidence that the forecasting framework satisfies core Human-Centred AI transparency requirements.

| Feature | SHAP (%, Rank) | LIME (%, Rank) | Permutation (%, Rank) |
|---|---|---|---|
| current_chan2 | 17.26 (1) | 12.80 (2) | 30.21 (2) |
| current_chan1 | 12.12 (3) | 11.95 (3) | 33.10 (1) |
| current_chan3 | 12.87 (2) | 12.93 (1) | 28.22 (3) |
| hour_sin | 8.08 (6) | 9.84 (4) | 3.74 (4) |
| hour_cos | 10.23 (4) | 8.83 (7) | 2.11 (5) |
| battery | 8.11 (5) | 9.65 (5) | 0.01 (10) |
| weekend | 6.22 (8) | 8.83 (7) | 1.68 (6) |
| hour | 7.55 (7) | 8.90 (6) | -0.40 (13) |
| signal | 6.22 (8) | 7.87 (9) | 0.89 (7) |
| dayofweek | 5.22 (10) | 8.40 (8) | 0.43 (8) |
| current_chan4 | 6.11 (9) | 0.00 (10) | 0.10 (9) |
| assumed_power_factor | 0.00 (11) | 0.00 (10) | -0.00 (11) |
| assumed_voltage_v | 0.00 (11) | 0.00 (10) | -0.08 (12) |

Table 5.1: Feature importance and ranking obtained from SHAP, LIME, and Permutation Importance. Values are reported as Importance (%). Rank.

## 5.2 Discussion

### 5.2.1 Addressing Research Questions

RQ1 asked to what extent an AI model could predict short-term commercial building energy consumption using high-resolution IoT sensor data. This was addressed by

developing, comparing, and optimising six deep learning forecasting architectures, with the Stacked LSTM selected as the final model due to its stable learning behaviour, small generalisation gap, and balanced complexity relative to model capacity (Section 4.1). The optimised model achieved an RMSE of 7.61 kW, an MAE of 5.05 kW, and an $R^2$ of 0.6216 on the held-out test set, explaining approximately 62% of the variation in energy consumption. These findings demonstrate that deep-learning-based forecasting approaches can effectively capture temporal relationships from high-frequency IoT sensor measurements and generate practically useful predictions for commercial-building energy management and generate reliable short-term predictions, despite the absence of environmental variables such as temperature and humidity in the dataset. Overall, the findings confirm the potential of AI-driven forecasting approaches for supporting commercial building energy management and operational decision-making.

RQ2 asked how XAI techniques such as SHAP and LIME could improve the transparency of the forecasting model and identify the features driving its predictions. This was answered by applying SHAP, LIME, and Permutation Feature Importance to the optimised Stacked LSTM model, all three of which converged on the same three predictors - `current_chan1`, `current_chan2`, and `current_chan3` - as dominant, together accounting for the majority of the ranked feature importance (Table 5.1). Temporal features such as `hour_sin` and `hour_cos` consistently followed as secondary contributors, reflecting daily consumption cycles. Rather than offering three isolated views, the techniques functioned as complementary checks on one another: SHAP quantified global and local feature contributions, LIME explained individual predictions, and Permutation Feature Importance measured the performance cost of removing each feature. Their agreement increases confidence that the model's transparency reflects genuine, physically meaningful relationships in the data rather than artefacts of a single method.

### 5.2.2 Human-Centred AI Perspective

A key principle of Human-Centred Artificial Intelligence (HCAI) is that AI systems should be transparent and interpretable to the people who use them [2, 3]. Although deep learning models such as LSTM networks can achieve strong forecasting performance, they are frequently criticised for operating as black-box systems, where users can observe predictions without understanding how they were produced. To address this limitation, the forecasting framework developed in this research incorporated three XAI techniques - SHAP, LIME, and Permutation Feature Importance - to provide explanations alongside forecasts rather than presenting predictions as unexplained numerical outputs.

Mapped against the ALTAI assessment framework [1], this research focuses specifically on the transparency principle. Transparency was supported through the use of SHAP, LIME, and Permutation Feature Importance, which revealed the factors influencing model predictions and made the forecasting process more understandable. Rather than presenting energy forecasts as unexplained outputs, the framework provided insight into the contribution of individual features, enabling users to better understand how predictions were generated. The consistent results obtained across all three explainability techniques further strengthened the transparency of the forecasting framework and reduced the black-box nature commonly associated with deep learning models (Table 5.1).

The framework also improves interpretability by translating complex model behaviour into terms that can be understood in relation to individual input features. While the internal mathematical operations of an LSTM network remain complex, XAI techniques provide interpretable representations of the factors associated with a prediction. Interpretability in this context does not require users to understand every aspect of the neural network architecture; instead, it enables them to identify the key variables that influenced a forecast and understand the direction and magnitude of those influences.

The use of multiple XAI techniques strengthens this further by providing complementary rather than redundant perspectives: SHAP offers both local and global insight into feature contributions, LIME provides local explanations for individual forecasts, and Permutation Feature Importance measures the performance cost of removing each feature entirely. Together, these methods give a more comprehensive account of the forecasting model's behaviour than any single technique could provide alone. The consistency of results across all three methods further demonstrated that the model relied primarily on the electrical current measurements identified as the most influential predictors of energy consumption, reducing the black-box nature of deep learning-based energy forecasting and aligning the framework with the ALTAI [1] transparency principle.

### 5.2.3 Practical Applications

The forecasting framework can support commercial warehouse energy management by providing short-term predictions of energy consumption using IoT sensor data. These forecasts can help facility managers monitor demand, identify unusual consumption patterns, and improve operational planning. Integrating explainability techniques enables users to understand the factors influencing forecast outcomes, supporting transparent, informed decision-making rather than reliance on opaque model predictions. Additionally, the ablation study in Section 4.4 showed that compact LSTM architectures performed comparably to larger ones, suggesting that the framework can be deployed efficiently in real-world industrial environments with limited computational requirements [2, 3, 4].

## 5.3 Self-Appraisal

One of the most significant challenges I faced during this project was data preprocessing. The dataset was obtained from an industry partner and consisted of real-world IoT sensor data, which required substantial effort to understand and prepare for analysis. A key challenge was identifying which data attributes were relevant for forecasting and determining how the raw sensor measurements should be transformed and structured for model development. The dataset contained complex and nested information, requiring careful exploration, cleaning, and feature engineering before meaningful analysis could be performed. This stage was particularly important because the quality of the input data directly influenced model performance. Even before implementing Explainable AI (XAI) techniques, considerable time was spent understanding the dataset and ensuring that the selected features accurately represented energy consumption patterns. Through this process, I strengthened my data analysis, problem-solving, and critical thinking skills, while gaining valuable experience working with real-world industry data that was far more complex than typical academic datasets.

# 6 Conclusion and Future Work

## 6.1 Conclusion

This research investigated the use of deep learning and Explainable Artificial Intelligence (XAI) techniques for forecasting short-term energy consumption in a commercial frozen food warehouse. Using real-world IoT sensor data, an LSTM-based forecasting framework was developed to predict future energy demand from historical electrical measurements [6, 7, 8], directly addressing RQ1 and RQ2 as set out in Section 1.3.

The research included data preprocessing, feature analysis, model development, hyperparameter tuning, explainability analysis, and model evaluation to create a forecasting system that was both accurate and interpretable. The results demonstrated that LSTM-based models are capable of effectively learning temporal patterns present in commercial building energy consumption data [10, 13, 14]. Following model optimisation, the selected Stacked LSTM architecture achieved reliable forecasting performance and successfully captured relationships between historical electrical measurements and future energy demand, answering RQ1. The findings confirmed that high-frequency IoT sensor measurements provide valuable information for short-term forecasting and can support practical energy management solutions.

A key contribution of this study was the integration of Explainable Artificial Intelligence into the forecasting framework, addressing RQ2. SHAP, LIME, and Permutation Feature Importance were applied to explain model predictions, and their agreement in identifying the electrical current channels as the dominant predictors (Section 5.1.6) provided confidence that the Stacked LSTM had learned physically meaningful relationships rather than spurious correlations [9, 10].

The research also aligned with Human-Centred AI principles, as discussed in Section 5.2.2, by pairing forecasts with supporting explanations rather than presenting predictions as unexplained outputs. This addressed the ALTAI transparency principle [1] introduced in Section 2.4 by providing explanations for model predictions through SHAP, LIME, and Permutation Feature Importance. [2, 3, 4, 5].

The results demonstrate that accurate energy forecasting and explainable AI can be combined within a single framework to support energy management in commercial buildings. The study contributes to the growing body of research on trustworthy and transparent AI for energy-management applications by demonstrating that forecasting accuracy and explainability can be effectively integrated within a single Human-Centred AI framework.

Despite the encouraging results, several limitations should be acknowledged. The study utilised data from a single commercial facility and did not incorporate contextual variables such as weather conditions, occupancy levels, or operational schedules. These factors may explain part of the unexplained variance observed in the forecasting results and could influence the generalisability of the proposed framework to other commercial-building environments.

## 6.2 Future Work

Several opportunities exist to extend this research. First, the framework could incorporate anomaly detection by comparing predicted and actual energy consumption values to automatically flag abnormal usage patterns, equipment faults, or operational inefficiencies, supporting proactive maintenance.

Second, future research could investigate transfer learning to improve model adaptability across different commercial buildings. Knowledge learned from one facility could be transferred to another, reducing the training data required and improving performance where historical data is limited, enhancing scalability and practical deployment.

Finally, future work could evaluate more advanced forecasting models, such as hybrid deep learning architectures, attention-based networks, and transformer models, to capture more complex temporal patterns. Incorporating additional contextual variables, such as weather conditions, occupancy information, and operational schedules, may further enhance predictive performance and provide more robust energy forecasts.

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